

Audrey Houghton (she/her)

 github.com/audreymhoughton |  [linkedin.com/in/audreyhoughton](https://www.linkedin.com/in/audreyhoughton) |  audreymhoughton@gmail.com
 +1-612-462-1481 |  Google Scholar

TECHNICAL SKILLS

Programming: Python, UNIX/Linux, GitHub, LaTeX, Docker, Singularity

Infrastructure & Data: AWS (S3), Ceph, SLURM, Kubernetes, Globus, Grafana, Prometheus, ETL pipelines, SQL, batch processing, data validation

Statistics & Modeling: logistic regression, Cox survival analysis, causal inference & discovery (PC, GES, FCI, BOSS), predictive modeling, GIS analysis, QGIS visualization, EHR integration, geocoding/FIPS recovery

Imaging & ML: neuroimaging/MRI segmentation, astrophysics photometric reduction, time-domain imaging, deep learning inference workloads and compute planning (e.g., nnU-Net)

Productivity & Collaboration: Jira, Linear, Kanban, Agile/Scrum, SOP documentation, Slack, Microsoft Teams, Google Workspace, Box, Salesforce API automation, compliant web scraping, GitHub Copilot

Additional Exposure: Matlab, R, IDL, JavaScript/Node.js, Go, Ansible/AWX, RedHat, ReadTheDocs, Google Apps Script, GCP, Terraform, Datadog, CI/CD (CircleCI, GitHub Actions)

EXPERIENCE

University of Minnesota – Department of Psychiatry & Behavioral Sciences

Dec 2025 - Present

Research Software Engineer III

Minneapolis, MN (Remote)

- Develop and maintain statistical and computational analysis pipelines for GIS-linked electronic health record data
- Build geospatial data-generation workflows in Python/Jupyter for county- and tract-level overlays, geocoding quality control, and FIPS recovery from external API sources
- Develop publication-grade causal inference and survival analysis workflows with cross-validation and robustness validation in Python
- Write and interpret results for publications and grant mechanisms, serving as a data and methods resource for an interdisciplinary research team

CoreWeave, Inc.

Apr 2024 - Sept 2024

Fleet Operations Engineer

Remote

- Provisioned and maintained thousands of NVIDIA GPU nodes in Kubernetes clusters using Ansible, AWX, Grafana, and Go, achieving onboarding timelines of 1–2 weeks with 6+ months average production uptime
- Diagnosed and resolved hardware/software issues across the node lifecycle, saving approximately \$250,000 per node repaired
- Partnered with 10+ teams—including networking, hardware, operations, and support—to streamline troubleshooting, accelerate incident resolution, and document SOPs and post-mortems, improving efficiency and reliability
- Identified points in the node lifecycle where nodes were lost, implementing solutions that saved the company millions in revenue

Masonic Institute of the Developing Brain

Jan 2021 - Apr 2024

Research Software Engineer II (promoted to III)

Minneapolis, MN (Remote)

- Developed Python wrappers leveraging AWS S3 storage and SLURM optimization to accelerate image processing up to 12x
- Engineered a containerized infant MRI segmentation application with deep learning models, achieving 600x speed-ups and +53% accuracy
- Built ETL-based pipelines to transfer and process petabytes of imaging data 10x faster than prior industry benchmarks, incorporating statistical error analysis and validation to ensure reliability and reproducibility
- Authored comprehensive documentation and curriculum, setting standards for reproducible neuroimaging data processing

Calyxt

Oct 2019 - Aug 2020

Laboratory Assistant & Data Specialist

Roseville, MN

- Developed Python scripts for comprehensive data tracking and analysis methods

The MINVERA Project

Jan 2015 - May 2017

Research Assistant

Missoula, MT

- Developed a Python-based photometric data reduction pipeline for detecting transiting exoplanets from time-domain imaging data, including calibration, aperture photometry, and light curve analysis

Rutgers University Astrophysics Department

May 2016 - Aug 2016

Research Assistant

New Brunswick, NJ

- Refactored IDL scripts into Python
- Generated models for galactic simulations through parameter optimization

PRODUCTS

Sales Summary Pipeline

Independent Automation Project

2025

[Proprietary]

- Built a proof-of-concept pipeline integrating Salesforce logging, CSV rotation, and Microsoft Teams alerts with compliant web scraping (robots.txt, throttling, retries)

CoreWeave Internal Documentation

CoreWeave, Inc.

2024

[Proprietary]

- Initiated and built company-wide documentation from the ground up, replacing scattered one-off documents with an organized, centralized Notion knowledge base covering terminology, SOPs, and team workflows

SLURM Wrappers

High-Performance Computing S3 and Disk SLURM-Compliant Wrappers

2021 - 2024

github.com/DCAN-Labs/SLURM_wrappers

- Bash/Python wrappers adaptable to any SLURM-based data pipeline

ABCD Study Collection 3165 Data Processing

NIH Data Archive (NDA)

2021 - 2024

collection3165.readthedocs.io

- Community dataset enabling access to petabytes of processed/unprocessed MRI data under BIDS standards

DCAN Labs Informational Guide

Minnesota Supercomputing Institute Data Engineering Documentation

2022 - 2024

dcan-labs-informational-guide.readthedocs.io

- Comprehensive, continually-supported documentation of usage of the Minnesota Supercomputing Institute to process, analyze, and transfer MRI data

CABINET

Container-Linking Wrapper

2022 - 2023

github.com/DCAN-Labs/CABINET

- Python-based universal container-linking wrapper employing FAIR principles

BIBSNet

Infant MRI Brain Segmentation Application

2022 - 2023

github.com/DCAN-Labs/BIBSnet

- Containerized infant MRI application; achieved 600x faster processing and +53% accuracy

EDUCATION

University of Montana - Missoula, MT

Bachelors of Arts in Physics

Aug 2013 - May 2017

umt.edu/physics-astronomy

University of Minnesota

Data Management Series; Project Management Certificate Coursework

2023